

Lab 1: Voltage and Current Sensors

Introduction

Simulation is a really important tool in engineering and science. Essentially, rather than actually building something in the real world, we can model the system using mathematics, and then use a computer to evaluate those equations to see how different parameter changes affect the system behaviour.

The objective of this lab is to use LTspice to model the AC source-load setup used in the project, understand its behaviour and then with this information develop the voltage as well as current sensors needed for the project. LTspice is a [SPICE](#)-based simulation tool that can help us analyse the behaviour of complex electrical circuits. LTspice can be downloaded for free from the [Analog Devices](#) website and is a tool widely used in industry for verifying design ideas. As with any simulation tool, the results given by LTspice are only as reliable as the details captured in your circuit model. Students should appreciate that it is extremely difficult to model the behaviour of real-life circuits accurately, and therefore should exercise caution when interpreting simulation results. For this reason, students must first use a combination of theory and practice to predict the behaviour of the circuit under study so that we can confirm to ourselves that the simulation results are actually correct. To help students get used to this form of thinking, since you will eventually have to think this way as a 'real life design engineer', all questions in this lab require students to derive theoretically predicted behaviour of each circuit before simulating. The simulation tool is then used to validate your theoretical and practical understanding of the circuit analysed. If your theory doesn't match with simulation results, then chances are that you have done something wrong either in your theoretical analysis or the simulation.

Read the entire lab manual before you start. This lab should take you approximately **6 hours**. **You should aim to complete Parts 1 and 2 during the 1st support session, and Parts 3 and 4 during the 2nd support session. To be able to do so, you will need to read this document carefully, before attending the 1st support session. The final answers should be written in this document, with supporting calculations shown in your logbook.** Before starting the lab, pull the lab repository from GitHub as it contains the LTspice model(s). Remember to commit and push your saved work to the lab repository on GitHub after completing each task.

Part 1: Revising Basics

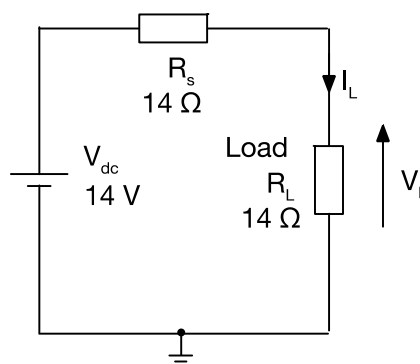
The first part of this lab intends to remind you of what you have learnt in ELECTENG 291. We will get you to use LTspice to do transient time simulation of a few very basic circuit models. Since you have already learnt how to use LTspice to draw a circuit model, we have provided you an LTspice file containing partially completed circuits that you could use for this lab. Here we use ideal device models as the devices you will be using are well within specifications and thus behave close to ideal devices. However, it should be noted that your simulation results are only as good as the underlying models used. As the device models used are not exact replicas of the real-world behaviour, simulation results always have errors and you must be mindful of these errors. If you must use detailed device models, LTspice contains a comprehensive library of device models. You can also

download 3rd party models or even create your own device models if required to improve the accuracy of your simulations.

As you learnt in the previous semester, before you can start simulating, you must first draw your circuit in the LTspice schematic editor window by adding components, wiring, and assigning values/part numbers. You must include a ground node, which is taken as the 0V reference node by the simulator. LTspice converts this graphical input given by you to something called a "Netlist", which is a text file that describes the interconnections between your components. SPICE then uses Nodal Analysis to work out all the voltages and currents in your circuit starting from 0s and stepping through the calculations in small "Timesteps" until the "Stop Time" you define under "Transient Analysis Command". Make sure to select a "Stop Time" that is sufficiently large so you can observe the steady-state behaviour of your circuit. But don't make the stop time very long as it will cause the simulation to run for a long time. Although LTspice tries to intelligently determine a suitable "Timestep", it may help in many occasions to define the "Maximum Timestep" to obtain accurate results. The coarser the "Timestep", the larger the error, but a really small "Timestep" can lead to a longer simulation time due to increased processing demand. As a designer, you must strike a balance and a good starting point is to select a "Timestep" that's about $1/10^{\text{th}}$ to $1/100^{\text{th}}$ of the period of the fastest frequency component in your circuit. In some cases, it also helps to define the initial conditions of your circuit to accelerate convergence of your simulation. LTspice by default tries to evaluate a suitable initial condition for your circuit, which in some cases may lead to inaccurate results.

After you "Run" the simulation, you can view voltage, current, instantaneous power, etc. waveforms in the waveform viewer by probing the circuit. You may also enter mathematical equations to format the data displayed in the waveform window. You can use cursors to measure salient features of your waveforms. You may also use the built-in capabilities of the waveform viewer to obtain RMS, Averages, and Integrals of the waveforms. However, note that these values are based on the waveforms displayed on the window and not having an exact number of cycles will lead to an incorrect reading. Remember to "Run" the simulation every time you make a change to the circuit. If you have forgotten how to use LTspice, then refer to the ELETENG 291 labs.

Q 1.1: We are going to analyse the simple circuit below to understand the basic functionality of LTspice:



First, using your theoretical understanding, determine the load current (I_L), load voltage (V_L), and power (P_L) delivered to the load resistor (R_L).

$$I_L: 0.5 \text{ A}$$

$$V_L: 7 \text{ V}$$

$$P_L: 3.5 \text{ W}$$

Q 1.2: After predicting how the circuit should behave theoretically, you should now simulate the circuit in LTspice and validate your theoretical analysis. Use the LTspice model provided to you (labelled Part 1A).

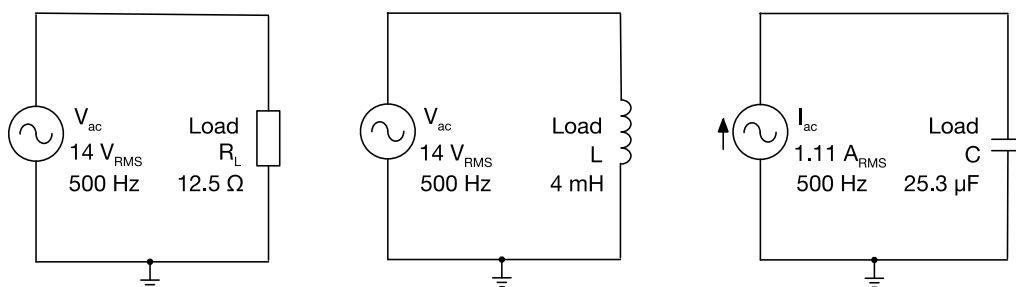
Parameter	Theoretical Value	Simulated Value
I_L	0.5 A	0.5 A
V_L	7 V	7 V
P_L	3.5 W	3.5 W

Comments (state if results match/differ):

Results are identical

Hopefully, in the above case, the output of LTspice matches your theoretical analysis.

Q 1.3: We are now going to extend our analysis into the following 3 AC circuits to understand the importance of timestep, initial conditions, and settling time (note that the circuit with the capacitor is driven by an AC current source):



What would be an appropriate "Maximum Timestep" and "Stop Time" to be used for this simulation?

Time Step (assuming 1/20th of the period of 500 Hz): 0.0001 s

Stop Time (assuming we'd like to capture 200 cycles): 0.4 s

Q1.1

$$I_L = V_{dc} / (R_s + R_L) = 14 / (14 + 14) = 1/2 \text{ A}$$

$$V_L = (R_L / (R_s + R_L)) \times V_{dc} = (1/2) \times 14 = 7 \text{ V}$$

$$P_L = V_L I_L = \frac{1}{2} \times 7 = 3.5 \text{ W}$$

Q1.3

$$\frac{1}{20} \times \frac{1}{500} = \frac{1}{10000} \text{ s} = 0.0001 \text{ s}$$

$$\frac{1}{500} \times 200 = 0.4 \text{ s}$$

If you simulate these 3 circuits, in their ideal form, you will observe that both the inductive and capacitive circuits will not reach a steady-state within the "Stop Time" you defined (i.e. 200 cycles). As a result, your simulation results for the inductive and capacitive loads will not agree with circuit theory, which would assume these circuits would have reached steady-state. This is because the inductor and capacitor are modelled to be ideal (i.e. resistance of the circuit is 0Ω) and as a result the time constant of these two circuits are equal to infinity. This is a common problem encountered when trying to simulate the behaviour of a real circuit. Therefore, it is common practice to add small resistive components in some branches of the circuit to help the simulation settle down. These resistors help reduce the time constant of the circuit and enable the simulations to reach steady-state faster. It should be noted that adding a resistive component changes the circuit behaviour and thus you should be careful to keep it to a comparatively small value to minimise the errors introduced.

In real life, both inductors and capacitors have some resistive component associated with them. For example, the winding of an inductor has some resistance, which is often referred to as the Equivalent Series Resistance (ESR) of an inductor. These resistive components associated with real circuit elements, help a real circuit reach a steady-state.

Q 1.4: In the simulation model provided (labelled Part 1B-1D), add a 0.2Ω resistor in series with the inductor and a 790Ω resistor in parallel with the capacitor to help the two circuits reach steady-state conditions faster. What is the time-constant of each circuit? Will your simulation reach steady-state well before the "Stop Time" you have defined? (i.e., is the "Stop Time" larger than 5 times the time constant?)

Time constant of circuit with 4mH inductor: 0.2 s

Time constant of circuit with $25.3\mu\text{F}$ capacitor: 0.1999 s

Would simulation reach steady-state:

Yes

Q 1.5: Calculate the theoretical load current ($I_{L(RMS)}$), load voltage ($V_{L(RMS)}$), peak instantaneous power ($P_{L(t)}$), and average power delivered by the source (P_{in}) for the circuits shown above. When calculating, ignore the impact of the resistive components you added in to the inductive and capacitive circuit. This is because we have chosen them not to significantly impact the behaviour of the each circuit. Simulate these circuits in LTspice using the models provided (labelled Part 1B-1D) and compare with the theoretical results (remember to modify these models as per Q1.4 above).

Q1.4: $\tau_I = \frac{L}{R} = \frac{4 \times 10^{-3}}{0.2} = 0.02 \text{ s}$

$$\tau_C = RC = 790 \times 25.3 \times 10^{-6} = 0.01999 \text{ s}$$

Recall: stop time = 0.4 s

$$5\tau_I = 0.02 \times 5 = 0.1 \text{ s}$$

$$5\tau_C = 0.01999 \times 5 = 0.09995 \text{ s}$$

$$0.4 \text{ s} > 5\tau_I \text{ and } 0.4 \text{ s} > 5\tau_C$$

$\therefore$ Simulation will reach steady state

Circuit with 12.5Ω Resistor

Parameter	Theoretical Value	Simulated Value
$I_{L(RMS)}$	1.12 A	1.12 A
$V_{L(RMS)}$	14 V	14 V
Peak $P_{L(t)}$	31.36 W	32 W
P_{in}	15.68 W	15.68 W

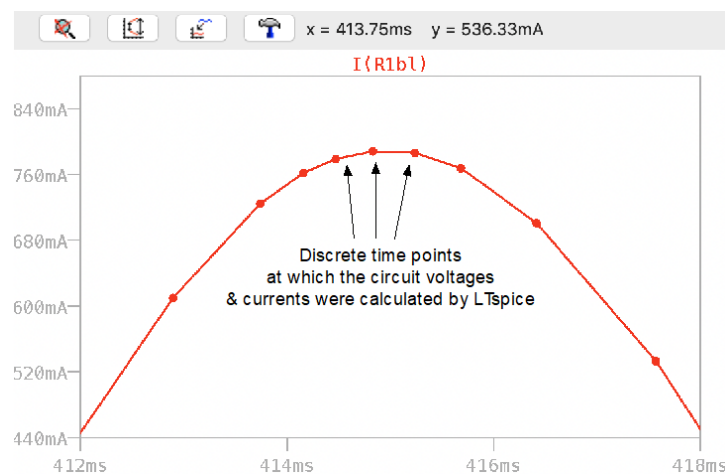
Circuit with 4mH Inductor

Parameter	Theoretical Value	Simulated Value
$I_{L(RMS)}$	1.114 A	1.0946 A
$V_{L(RMS)}$	14 V	14.006 V
Peak $P_{L(t)}$	15.59 W	15.4 W
P_{in}	0 W	-21.23 mW

Circuit with 25.3μF Capacitor

Parameter	Theoretical Value	Simulated Value
$I_{L(RMS)}$	1.11 A	1.11 A
$V_{L(RMS)}$	13.96 V	13.92 V
Peak $P_{L(t)}$	15.5 W	15.45 W
P_{in}	0	480 mW

Q 1.6: Your simulated results for the 12.5Ω resistive load will be somewhat different to what you predicted from circuit theory. Zoom to observe simulated waveforms near the peaks and you will see granularity of the waveform shape caused by the large timestep:



Q1.5)

Resistor:

$$I_{L,RMS} = \frac{14}{12.5} = 1.12 \text{ A}$$

$$V_{L,RMS} = 14 \text{ V (from LTSPICE file)}$$

$$\text{Peak } P_L = (I_{L,RMS} \times V_{L,RMS}) \times 2 = (1.12 \times 14) \times 2 = 31.36 \text{ W}$$

$$P_{in} = I_{L,RMS} \times V_{L,RMS} = 1.12 \times 14 = 15.68 \text{ W}$$

Inductor:

$$\begin{aligned} \text{Inductive impedance } X_L &= 2\pi fL \\ &= 2\pi 500 \times 4 \times 10^{-3} = 4\pi \Omega = 12.57 \Omega \end{aligned}$$

$$I_{L,RMS} = \frac{V_{L,RMS}}{X_L} = \frac{14}{4\pi} = 1.114 \text{ A}$$

$$V_{L,RMS} = 14 \text{ V}$$

$$\text{Peak } P_L = I_L^2 \cdot X_L = 1.114^2 \times 4\pi = 15.59 \text{ W}$$

$$P_{in} = 0 \text{ W (fully inductive)}$$

Capacitor:

$$\text{Capacitive Impedance } X_C = \frac{1}{2\pi f C}$$

$$= \frac{1}{2\pi 500 \times 25.3 \times 10^{-6}}$$

$$= 12.58 \Omega$$

$$I_{L, \text{RMS}} = 1.11 \text{ A (given by LTspice file)}$$

$$V_{L, \text{RMS}} = 1.11 \times 12.58 = 13.97 \text{ V}$$

$$P_L \text{ Peak} = I_L^2 \cdot X_C = 1.11^2 \times 12.58 = 15.5 \text{ W}$$

$$P_{\text{in}} = 0 \text{ W (fully capacitive)}$$

How can we minimise the granularity and improve the accuracy? What would be a drawback of doing so? Simulate the resistive load by setting the timestep to a finer value and state your observations.

You can minimise the granularity by decreasing the size of the timestep, however the simulation takes a long time to load.

Q 1.7: Modify the simulation by removing the resistive components you added as per Q1.4 and run it. Compare the simulation results you obtained from the modified circuits, which now represent an ideal inductive load and an ideal capacitive load, to show how the resistors help these circuits reach steady-state. State your observations.

Simulator will not run as it cannot reach a steady state.

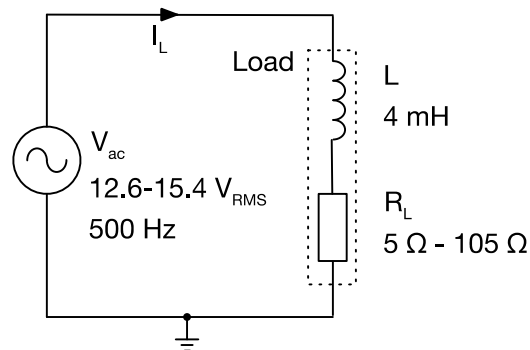
From the above tasks, you should realise that a simulation package may simulate even some of the simplest circuits incorrectly if not set up properly. Therefore, as a design engineer, you must learn to appreciate the limitations of your simulation tools and always use first-principles to understand the behaviour of your circuit (even if it is a very complex circuit) before simulating. Theoretical understanding of the circuit behaviour will help you choose a suitable simulator and set up appropriate simulation conditions.

Part 2: Modelling Your AC Load

Now that you have some familiarity with LTspice, you can use it to help speed up the development of your analogue circuitry required for the energy monitor. The first thing you should aim to do is to model the AC load you will be using in LTspice.

Having a model of the AC load will help you investigate how the load, as well as your analogue circuitry, behaves under all operating conditions.

The AC load we will use in the project compose of a series connected inductor and a variable resistor. The inductor has a nominal inductance of about 4mH. The inductor also has some internal series resistance (ESR), which is about 5Ω. Depending on the position of the viper, the variable resistor can have a nominal resistance between 0Ω and 100Ω. Thus, your nominal (or average) load can be modelled as a series combination of a 4mH inductor (L) and a variable resistance (R_L) between 5Ω and 105Ω, as shown by the circuit:



It should be noted that, as with any practical device, we should expect some variation in the inductance of the inductor as well the resistance range of the variable resistor. This can be due to, for example, manufacturing tolerances. This means that, we can expect each AC load to have a slightly different inductance and resistance range.

As indicated in the circuit above, the voltage of the AC source (V_{ac}) can change between 12.6V_{rms} and 15.4V_{rms}. This comes from the design specifications given in the course outline, which states the supply voltage can change by $\pm 10\%$. Though the supply frequency can vary a little as per the specifications, we are going to ignore this, as the impact is somewhat negligible. So, we are going to assume that the frequency of the AC voltage is 500Hz.

When V_{ac} is a maximum (15.4V_{rms}) and R_L is a minimum (5Ω), the AC load will draw maximum VA, and you can calculate this as 17.5VA (taking the inductance as 4mH). When V_{ac} is a minimum (12.6V_{rms}) and R_L is a maximum (105Ω), the AC load will draw minimum VA, and you can calculate this as 1.5VA. However, the design specifications given to you state the maximum and minimum VA your design should be capable of measuring is limited to 7.5VA and 2.5VA, respectively. When we test your design, we will make sure to set the load to be within this range by setting R_L and V_{ac} to appropriate values. For example, we may set V_{ac} to 12.6V_{rms} and adjust R_L so that the AC load draws 7.5VA. Or we may set V_{ac} to 15.4V_{rms} and adjust R_L so that the AC load draws 2.5VA. Or we may set V_{ac} and R_L to any other combination resulting in 2.5VA-7.5VA. Also note, though we assumed the inductive part of the load is 4mH, this value can vary quite a bit due to manufacturing tolerances (as much as 50%). Hence the reason why your design specifications also state a range of power factors you can expect to see.

Q 2.1: As stated in your design specifications, the AC voltage supplied by the power supply across the AC load is going to be between 12.6V_{rms} and 15.4V_{rms}. The AC load is going to be set to draw 2.5VA-7.5VA regardless of the AC voltage

supplied to it. What is the maximum and minimum current you expect to measure? What is the maximum and minimum AC voltage you expect to measure?

Maximum RMS load voltage (given by your design specs): 15.4 V
 Minimum RMS load voltage (given by your design specs): 12.6 V
 Maximum RMS load current: 0.596 A
 Minimum RMS load current: 0.162 A

Q 2.2: Assume that the wiper on the variable resistor is set somewhat closer to the lowest resistance position, and therefore R_L is 25Ω . We are also assuming that L is 4mH and V_{ac} is $14V_{rms}$ in this specific example. Note that these numbers are chosen as an example load setting we may test your circuit at. Using circuit theory, determine the RMS load current as well as the real, reactive, and apparent power consumed by the load. Verify answers using the LTspice model provided (labelled Part 2).

Parameter	Theoretical Value	Simulated Value
Load current ($I_{L(rms)}$)	0.5 A	0.5 A
Real Power (W)	6.26 W	6.257 W
Reactive Power (VAR)	3.147 VAR	3.166 VAR
Apparent Power (VA)	7.006 VA	6.7 VA

Comment (state if results match/differ and why):

Results mostly match

Q 2.3: Similarly, calculate and simulate the RMS load current, real, reactive, and apparent power if the wiper position is changed to increase R_L to 75Ω (i.e. closer to the highest resistance position)? Again, these numbers are chosen as an example setting we may test your circuit at (note that L is 4mH and V_{ac} is $14V_{rms}$).

Parameter	Theoretical Value	Simulated Value
Load current ($I_{L(rms)}$)	0.1841 A	0.184 A
Real Power (W)	2.592 W	2.5925 W
Reactive Power (VAR)	0.4259 VAR	0.426 VAR
Apparent Power (VA)	2.577 VA	2.55 VA

Comments (state if results match/differ and why):

Results mostly match

$$Q2.1: \quad \frac{7.5 \text{ VA}}{12.6 \text{ V}} = 0.596 \text{ A}$$

$$\frac{2.5 \text{ VA}}{15.4 \text{ V}} = 0.162 \text{ A}$$

$$Q2.2: \quad R_L = 25 \Omega, \quad L = 4 \text{ mH}, \quad V_{\text{rms}} = 14 \text{ V}:$$

$$X_L = 2\pi f L = 2\pi \times 500 \times 4 \times 10^{-3} = 4\pi = 12.57 \Omega$$

$$Z_{\text{total}} = \sqrt{25^2 + 12.57^2} = 27.98 \Omega$$

$$I_{L, \text{rms}} = \frac{V_{\text{rms}}}{Z_{\text{total}}} = \frac{14}{27.98} = 0.5004 \text{ A}$$

$$\theta = \tan^{-1}\left(\frac{12.57}{25}\right) = 26.69^\circ$$

$$\text{Apparant power: } S = V_{\text{rms}} I_{\text{rms}} = 14 \times 0.5004 = 7.006 \text{ VA}$$

$$\text{Real power: } S \times \cos \theta = 7.006 \times \cos(26.69^\circ) = 6.26 \text{ W}$$

$$\text{Reactive power} = S \cdot \sin \theta = 7.006 \times \sin(26.69^\circ) = 3.147 \text{ VAR}$$

Q 2.3: $R_L = 75 \Omega$, $L = 4 \text{ mH}$, $X_L = 12.57 \Omega$ (same as before)

$$Z_t = \sqrt{75^2 + 12.57^2}$$

$$= 76.05 \Omega$$

$$I_{L \text{ RMS}} = \frac{V_{\text{RMS}}}{Z_t} = \frac{14}{76.05} = 0.1841 \text{ A}$$

$$\theta = \tan^{-1} \left(\frac{12.57}{75} \right) = 9.514^\circ$$

$$\text{Apparent power} = S = V_{\text{RMS}} \times I_{\text{RMS}} = 2.577 \text{ VA}$$

$$\begin{aligned} \text{Real power} &= S \cdot \cos \theta = 2.577 \times \cos(9.514^\circ) \\ &= 2.542 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Reactive power} &= S \cdot \sin \theta = 2.577 \times \sin(9.514^\circ) \\ &= 0.4259 \text{ VAR} \end{aligned}$$